

Learning to summarize from human feedback*

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Abstract—This paper implements and extends the approach proposed in “Learning to Summarize from Human Feedback” (Stiennon et al., 2020), focusing on training summarization models to align with human preferences instead of traditional metrics like ROUGE. Our work comprises two main phases: first, generating a preference-labeled dataset using the BART model and ROUGE-based selection; second, training a reward model based on this dataset to emulate human judgment. This pipeline serves as a foundation for reinforcement learning optimization using Proximal Policy Optimization (PPO), although that stage remains unimplemented due to resource constraints. The framework demonstrates scalable alternatives to manual evaluation and sets the stage for future extensions.

Index Terms—between 2-5 keywords

I. INTRODUCTION

Summarization models often optimize for ROUGE or similar metrics, which do not necessarily correlate with human preferences. This project aims to realign model training with human judgment, following the principles laid out by Stiennon et al. Our work includes the creation of a preference-labeled dataset, the training of a reward model that predicts human-preferred outputs, and analysis of generalizability. This method promotes ethical AI alignment and improves real-world performance of summarizers.

II. RELATED WORK

Prior summarization research has focused on supervised learning using ground-truth summaries and evaluation using ROUGE. While effective, these metrics fail to capture subjective human judgment. RLHF methods introduced by Stiennon et al. use direct human feedback to fine-tune models. Our approach extends this by automating parts of feedback collection and modeling, creating a more scalable framework.

III. METHODOLOGY

A. Phase 1: Supervised Summarization

We fine-tuned BART on Reddit TL;DR data using greedy and beam search decoding. Preference labels were derived by comparing ROUGE scores and selecting the higher-scoring summary.

B. Phase 2: Reward Model Training

We trained a BERT-based classifier to predict preferred summaries based on article-summary pairs. This binary classifier mimics human evaluation.

C. Phase 3 (Planned): Reinforcement Learning

Although not implemented, the next phase would involve optimizing the summarization model using PPO with the reward model as feedback.

D. Phase 4: Tabular data and images

Although not implemented, the next phase would involve optimizing the summarization model using PPO with the reward model as feedback.

IV. EXPERIMENTAL SETUP

Dataset: Reddit TL;DR corpus.

Metrics: ROUGE-L for initial preference generation; classifier accuracy for reward model.

Tools: Hugging Face Transformers; training on GPU-based environments.

Hyperparameters: Standard BERT training with cross-entropy loss.

V. RESULTS AND DISCUSSION

The classifier successfully learned to distinguish high-ROUGE summaries, indicating effectiveness of proxy-labeled preferences. While lacking true human labels, the results suggest that scalable feedback can simulate human-like evaluation. Limitations include the risk of overfitting and absence of true human feedback validation.

VI. CONCLUSION

This paper replicates and extends RLHF summarization methodology, contributing a scalable reward modeling pipeline. Future work includes implementing PPO fine-tuning, integrating human evaluators, and expanding to multilingual and multimodal summarization.

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